Early signals of ejection in the three-body problem: the role of mass ratios

Willian Wistuba Melo da Cunha
Self-taught physics enthusiast

We ask whether the binary that survives the disruption of a three-body system can be recognized before the third body is ejected. In 7,000 regularized planar integrations, starting from rest and with virial ratio 0.5, we compare equal masses, two small and one large, and two large and one small. With equal masses there is no early signal: the final pair only becomes definitively the most bound at the last close encounter, and the predictor based on pair energies is correct 35% to 45% of the time, close to chance. With unequal masses, the lightest body is ejected in 92% to 99.8% of the cases, in agreement with the approximate m^{-3} law, and the signal grows with the disparity: with masses 10, 10, and 1, the final pair is determined at half the time preceding the ejection. Even so, the rule that the two heaviest bodies remain predicts the outcome as well as or better than any dynamical signal. Two large masses eject the small one earlier and with fewer escape attempts.

Keywords: three-body problem; ejection; final binary; mass ratio; predictability; regularization.

ORCID: 0009-0008-2875-6607

Published: 3 November 2026. Version 2. Not peer reviewed.

Series: The three-body problem, part 5 of 8. https://fisica-revisitada.vercel.app/en/articles/sinais-da-ejecao/

Translated from the original in Brazilian Portuguese.

I. INTRODUCTION

In the nonhierarchical regime, the three-body problem almost always ends in the same way: two bodies form a binary and the third is ejected [1, 2]. The identity of the ejected body and the properties of the final binary admit a statistical description [2, 3], but the individual trajectory is chaotic. This note asks a more specific question: is it possible to recognize, in the trajectories, the binary that will survive before the third body is ejected? If so, how far in advance, and does this depend on the mass ratios?

We test three classes of masses: three equal masses, two small and one large, and two large and one small. We also include a scaling control, with the equal masses multiplied by ten, to verify that, in Newtonian gravity, a uniform change of mass only alters the time scale.

II. METHOD

A. Initial conditions and ensembles

The simulations are planar, in units with $G = 1$. The three initial positions are drawn uniformly in the unit disk, rejecting pairs closer than 0.1, and the center of mass is moved to the origin. We use two ensembles: start from rest (cold collapse) and start with random velocities rescaled to virial ratio $K/|W| = 0.5$. The mass groups are (1, 1, 1), (10, 10, 10), (1, 1, 3), (1, 1, 10), (3, 3, 1), and (10, 10, 1), with 500 runs per group and per ensemble. The (10, 10, 10) case was run twice: with the same seeds as the (1, 1, 1) case and with independent seeds. Time is measured in units of the crossing time $t_{\rm cr} = GM^{5/2}/(2|E|)^{3/2}$, where M is the total mass and E the energy.

B. Integration and ejection criterion

The integration uses algorithmic regularization in chain coordinates [4, 5], with Gragg-Bulirsch-Stoer extrapolation and tolerance 10^{-13}, without softening of the potential. The relative energy error had a median of about 3×10^{-13} and a maximum of 2×10^{-9}; no run was discarded. A body k is considered ejected when the pair formed by the other two is bound, the energy of k relative to the center of mass of the pair is positive, k is receding, and its distance exceeds 20 times the semimajor axis of the pair. The ejection time $t_{\rm ej}$ is that of the last pericenter of k before escape. Runs without ejection up to $2000\,t_{\rm cr}$ are recorded as unresolved.

C. Measured signals

Along each trajectory we compute the pair energies

$$E_{ij} = \tfrac{1}{2}\mu_{ij} v_{ij}^2 - \frac{G m_i m_j}{r_{ij}}, \qquad \mu_{ij} = \frac{m_i m_j}{m_i + m_j}, \tag{1}$$

and call the most bound pair the one with the lowest E_{ij}. We define $t_{\rm lock}$ as the last instant at which the most bound pair becomes the final pair, and the lead time as $t_{\rm ej} - t_{\rm lock}$. We evaluate the predictor "the final pair is the currently most bound pair" at fractions $f = 0.25$, 0.5, 0.75, and 0.9 of $t_{\rm ej}$, and compare it with the naive predictor "the two heaviest bodies remain" and with chance (1/3). An escape attempt is an excursion of the body that will be ejected beyond five semimajor axes of the final binary, while still bound, followed by a return.

III. RESULTS

A. Mass scaling

The (1, 1, 1) and (10, 10, 10) cases are statistically indistinguishable when time is measured in t_{cr}: in the Kolmogorov-Smirnov test, $p \approx 0.9998$ for the distribution of t_{ej}/t_{cr} with the same seeds, and $p = 0.94$ and 0.67 with independent seeds, in the two ensembles. In absolute time, the more massive case evolves about $10^{1/2}$ times faster, as predicted by the scaling $t \propto M^{-1/2}$. With the same seeds, the runs that end before $10\,t_{cr}$ are identical; in the longer ones, the outcome begins to diverge, because rounding differences are amplified by chaos.

B. Which body is ejected

Table I summarizes the results. With unequal masses, the ejected body is almost always one of the lightest: from 92% to 99.8% of the ejections, in agreement with the approximate prediction of a probability proportional to m^{-3} [1].

C. Lead time and predictors

With equal masses, there is no early signal: the median lead time is practically zero, which means that the final pair only becomes definitively the most bound at the last close encounter, and the most bound pair predictor is correct 35% to 45% of the time at any fraction of t_{ej}, slightly above chance. Fig. 1 compares the predictors for the start from rest. The signal grows with the mass disparity and is largest in the (10, 10, 1) case, in which the final pair is already formed, in the median, for about half (rest) to four fifths (virial) of the time preceding the ejection. In all groups with unequal masses, however, the naive predictor, based only on the masses, is as good as or better than the dynamical predictor. In the (1, 1, 3) case, the naive predictor cannot distinguish the two light masses and stays near 50%.

D. Escape attempts and final binary

With equal masses, there are on average 10 to 12 escape attempts before the ejection. With two large masses and one small, the ejection is faster and more direct: fewer than two attempts on average for the start from rest. The difference is not explained by the shorter duration alone: the median rate of attempts per crossing time falls from 0.17 (equal masses) to 0.03 (3, 3, 1) and zero (10, 10, 1). The binding energies of the final pair increase in just over half of the transitions between consecutive attempts (54% to 76%), only a weak stepwise tightening. The final binaries are highly eccentric, with mean eccentricity between 0.68 and 0.99.

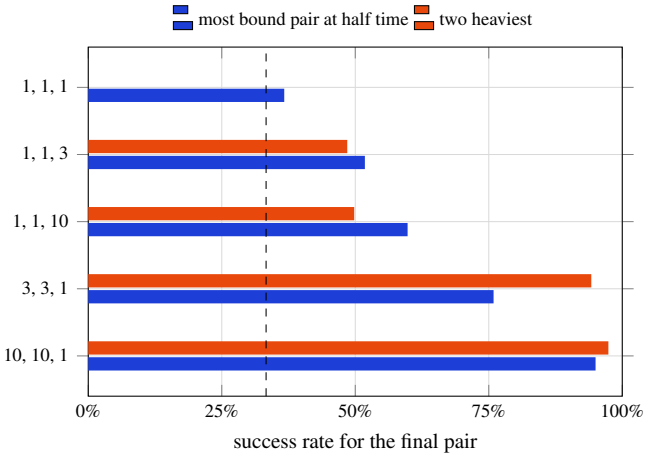


Fig. 1. Success rate for the final pair, for the start from rest, by mass group: most bound pair predictor evaluated at half the time to ejection, naive predictor of the two heaviest bodies (undefined for equal masses), and chance (dashed line).

IV. DISCUSSION

Conservation of momentum explains the asymmetry. In a close encounter, the bodies exchange equal amounts of momentum, and for the small mass this represents a large change in velocity. With two large masses, a single encounter is usually enough to throw the small one out, and the identity of the final pair is decided early. With equal masses, no body has this disadvantage, and the system exchanges partners several times before the outcome, so that the information about the final binary only appears at the last encounter. This is consistent with Heggie's law [6]: encounters with hard binaries tend to tighten them and to fling out the intruder.

The finding that the masses predict better than any energy signal indicates that, in the chaotic regime, the information useful for the outcome lies more in the global conditions (masses and energy) than in the details of the trajectory, which is the premise of the statistical solutions [2, 3].

The main limitations are the following. The simulations are planar. For the start from rest, the total angular momentum is zero, which forces highly eccentric final binaries; the mean of 0.99 in the (10, 10, 1) case is an effect of this initial condition. In the virial ensemble, 3% to 12% of the runs end with a direct ejection, with no prior pericenter, which dilutes the success rates. The definitions of most bound pair, of t_{ej}, and of escape attempt are our own choices. With 500 runs per row, the binomial error of the fractions is about two percentage points. Finally, as an independent check, we redid a sample of short runs with another integrator, with adaptive step size (DOP853), and with separately written code: in 16 of 17 runs the ejected body and the ejection time agreed; in the remaining, longer run the outcomes diverged, as expected for a chaotic system.

Table I. Results by mass group and by ensemble (R: start from rest; V: virial). Between 465 and 499 ejections per row, out of 500 runs. "Lightest" is the fraction of ejections of the lightest body, with the m^{-3} prediction in parentheses. Lead time: median of $(t_{\rm ej} - t_{\rm lock})/t_{\rm ej}$. Success rate: most bound pair predictor at $f = 0.5$ and naive predictor.

Masses	Ens.	Lightest	$t_{\rm ej}/t_{\rm cr}$	Lead time	Most bound pair	Two heaviest	Attempts	$\bar{e}$
1, 1, 1	R	no effect	28.9	0.000	0.37	chance	10.5	0.72
1, 1, 1	V	no effect	33.2	0.000	0.44	chance	11.6	0.84
1, 1, 3	R	0.970 (0.982)	7.9	0.001	0.52	0.49	4.6	0.77
1, 1, 3	V	0.970 (0.982)	17.3	0.007	0.53	0.49	8.7	0.80
1, 1, 10	R	0.996 (1.000)	7.4	0.032	0.60	0.50	7.6	0.78
1, 1, 10	V	0.998 (1.000)	10.0	0.105	0.67	0.50	13.1	0.78
3, 3, 1	R	0.942 (0.931)	5.9	0.020	0.76	0.94	1.9	0.87
3, 3, 1	V	0.916 (0.931)	12.4	0.015	0.72	0.92	3.1	0.77
10, 10, 1	R	0.974 (0.998)	2.5	0.455	0.95	0.97	0.6	0.99
10, 10, 1	V	0.989 (0.998)	3.6	0.802	0.95	0.99	2.2	0.68

V. CONCLUSIONS

With three equal masses, we find no early signal of the surviving binary: it is only determined at the last close encounter. With unequal masses, the signal exists and grows with the disparity, but it is dominated by a simple rule, that the two heaviest bodies remain together. Two large masses eject the small one earlier and with fewer attempts, not more often. A natural next step is to test whether the ratio between the orbital velocity of the pair and the approach velocity of the third body predicts the outcome better than the masses.

TRANSPARENCY AND USE OF ARTIFICIAL INTELLIGENCE

The author is a lawyer and not a physicist. This work results from independent research, conducted with the assistance of artificial intelligence (a language model) in the literature searches, the programming of the simulations, the analysis of the results, the writing, and the translation. The author formulated the questions, followed the experiments, and is responsible for the published content.

DATA AND CODE AVAILABILITY

All numbers were obtained with scripts published at https://fisica-revisitada.vercel.app/en/code/, which run on Node.js without external dependencies and allow the results to be reproduced.

CONFLICT OF INTEREST

The author declares no conflicts of interest. This work received no funding. This text has not undergone peer review and should be read as a working note. Corrections are recorded in the note itself, with the date.

CITE AS

W. W. M. da Cunha, "Early signals of ejection in the three-body problem: the role of mass ratios," Física Revisitada, Technical Note FR-TC-05 (2026), https://fisica-revisitada.vercel.app/en/articles/sinais-da-ejecao/.

[1] M. Valtonen and H. Karttunen, The Three-Body Problem (Cambridge University Press, Cambridge, 2006). doi.org/10.1017/CBO9780511616006

[2] N. C. Stone and N. W. C. Leigh, A statistical solution to the chaotic, non-hierarchical three-body problem, Nature 576, 406 (2019). doi.org/10.1038/s41586-019-1833-8

[3] J. J. Monaghan, A statistical theory of the disruption of three-body systems. I. Low angular momentum, Mon. Not. R. Astron. Soc. 176, 63 (1976). doi.org/10.1093/mnras/176.1.63

[4] S. Mikkola and S. J. Aarseth, An implementation of N-body chain regularization, Celest. Mech. Dyn. Astron. 57, 439 (1993). doi.org/10.1007/BF00695714

[5] S. Mikkola and K. Tanikawa, Algorithmic regularization of the few-body problem, Mon. Not. R. Astron. Soc. 310, 745 (1999). doi.org/10.1046/j.1365-8711.1999.02982.x

[6] D. C. Heggie, Binary evolution in stellar dynamics, Mon. Not. R. Astron. Soc. 173, 729 (1975). doi.org/10.1093/mnras/173.3.729